DFM Anonymisation

The aim of this project is to evaluate and pipeline methods for PII (personally identifiable information) detection and removal in plain text. So far the main focus has been on the *Danish* language, though much should generalize to other scandinavian/germanic languages.

Quick start

See README.md

Data and ground truth

Public Danish PII datasets does not seem to be in abundance. The ground truth so far used for evaluation is:

- DaNE
  - Only NER tags ORG, LOC, and PER are used, other categories are deemed not relevant in PII context
- Dansk-NER
  - Only NER tag PERSON is used, other categories are deemed not relevant in PII context
- Generated data using the Presidio library with the categories (160-170 sentences for each category):
  - ADDRESS
  - CARD EXPIRY
  - CARD NUMBER
  - CPR NUMBER
  - CVV NUMBER
  - DATE OF BIRTH
  - DRIVERS LICENSE
  - EMAIL
  - ORGANIZATION
  - PERSON
  - PHONE NUMBER
  - WEBSITE

Models

As of now we still have too little annotated data for actual (post) training of NER models specifically for PII. Therefore we so far only use off-the-shelf models like GliNER models or NER models with some overlap in categories with PII.

These are the models we looked at so far:

- scandiner
  - Using categories ORG, LOC, PER
- presidio
  - the standard presidio NER model for Danish. Based on the spacy model *da_core_news_trf*
- gliner-PII
- gliner-multi-v2.1

The model used by default by the CLI tool (`dfm-anonymization`) is an ensemble model based on gliner-pii and presidio. This is chosen since these two together are close to best in every category, while introducing more models would hurt performance.

Evaluation results

We evaluate per category and per model whenever both dataset and model include this category, then average the scores for each dataset (and then each category). Finally we rank based on the f5 score, weighing recall higher than precision.

model	precision	f1	f5	recall
ensemble	0.965112	0.945467	0.932161	0.931222
scandiner	0.960480	0.934935	0.915107	0.913586
gliner-pii	0.876750	0.851719	0.854775	0.856366
gliner_multi-v2.1	0.733916	0.778923	0.820998	0.722963
presidio	0.967630	0.818236	0.769630	0.766570

Full results for each category:

label	model	precision	f1	f5	recall
card	gliner-pii	1.000000	0.730123	0.584511	0.574956
time	gliner-pii	0.936940	0.967381	0.997408	1.000000
	gliner_multi	0.959983	0.977574	0.994465	0.995902
	presidio	0.984375	0.330753	0.205657	0.199378
drivers license	gliner-pii	1.000000	1.000000	1.000000	1.000000
	gliner_multi	0.883041	0.909639	0.935653	0.937888
email	gliner-pii	0.997854	0.983087	0.969838	0.968750
	presidio	1.000000	0.965517	0.935733	0.933333
	gliner_multi	0.795699	0.258290	0.159100	0.154167
iban	gliner-pii	0.987730	0.975758	0.964961	0.964072
	gliner_multi	0.549587	0.650367	0.782884	0.796407
loc	presidio	0.971716	0.944454	0.921885	0.920105
	scandiner	0.983142	0.919632	0.871915	0.868313
	gliner_multi	0.857555	0.812221	0.777663	0.775044
	gliner-pii	0.799862	0.691106	0.636982	0.633698
org	scandiner	0.983658	0.963033	0.946002	0.944662
	presidio	0.990338	0.957231	0.930327	0.928224
	gliner_multi	0.705424	0.761888	0.882310	0.898388
	gliner-pii	0.695549	0.740441	0.853925	0.869835
per	scandiner	0.929921	0.926404	0.923304	0.923051
	presidio	0.906235	0.913072	0.920433	0.921096
	gliner_multi	0.753707	0.805625	0.875209	0.882343
	gliner-pii	0.747345	0.777887	0.819428	0.823601
phone	gliner_multi	0.979920	0.989858	0.999212	1.000000
	gliner-pii	1.000000	0.993814	0.988172	0.987705
	presidio	1.000000	0.786070	0.656440	0.647541
ssn	gliner-pii	0.935673	0.958084	0.979746	0.981595
	gliner_multi	0.493450	0.576531	0.682621	0.693252
url	presidio	1.000000	0.863158	0.766355	0.759259
	gliner-pii	1.000000	0.855124	0.754256	0.746914